Semester VI **B. Tech CSE (AI)**

BCSAI 606: BLOCKCHAIN

2L + 0T + 0P + 2C **MM 100**

Unit 1:Introduction to Cryptography

Introduction to Advanced Cryptography and Cryptanalysis, Classical Encryption Techniques – Substitution Techniques, Transposition Techniques, Permutation Method. Advanced Encryption Techniques and Security Issues – RC4, One-time Pad, RSA, DES, Triple DES, AES and Diffie Hellman, Case study.

Unit 2: Conventional Encryption

Confidentiality using conventional encryption – Placement of Encryption, Traffic Confidentiality, Key Distribution and Random Number Generation.

Unit 3: Key management – Generating Keys, Nonlinear Keyspaces, Transferring Keys, Verifying Keys, Using Keys, Updating Keys, Storing keys, Backup keys, Compromised Keys, Lifetime of Keys, Destroying Keys and Public-Key Management, Case study.

Unit 4: Introduction to Blockchain and Crypto-currency Basics

What is Blockchain, Blockchain Technology and Mechanisms, Challenges, Centralized Servers and Trusted Third Party, Shift from gold standard to flat currency to Hash cash/Digital, Trust less System, Transactions and Blocks, Digital Signatures

Discussion on Bitcoin and Ethereum, Significance, Security, The Bitcoin Mining Network, Mining Developments, Decentralization and Hard Forks, Ethereum Eco-System

Unit 5: Working of Blockchain and Hyper Ledger

Technology behind Blockchain-Consensus Building, Proof of Work, Byzantine Generals. Distributed Consensus, Cryptography, Hashing, Data Integrity, Public vs. Private Key Cryptography, Merkle Trees. Crypto-currency and Mining, Proof of Work vs. Stake, Business Model.
What is Hyper Ledger, Distributed Ledger Technology, Hyper Ledger Fabric and Composer, Assets, Chaincode and Ledger/Transactions, Permission Network, Member Services, Nodes and Channels, Development Machine Specifications and IDE.

Text/ Reference Books:

1. Imran Bashir. (2018). Mastering Blockchain: Distributed Ledger Technology, Decentralization and Smart Contracts explained, Import.
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder. (2016). Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction. Princeton University Press.
3. Alex Tapscott and Don Tapscott. (2016). Blockchain Revolution: How the Technology behind Bitcoin Is Changing Money, Business, and the World, Portfolio.
4. Dr. Gavin Wood. (2014) Ethereum: A Secure Decentralized Transaction Ledger. Yellow Paper.
5. Chris Dannen. (2017). Introducing Ethereum and Solidity: Foundations of Cryptocurrency and Blockchain Programming for Beginners. Apress.
6. William Mougayar, VitalikButerin. (2016). The Business Blockchain: Promise, Practice and Application of the Next Internet Technology. Wiley.